
CS204: Algorithms & Data Structures Lab

Week # 11 (1 Questions, 157 Marks)

Lab session: AL1

Held on: 14-Oct-2024 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 2

Submission time: 16:45 Hrs, 14-Oct-2024

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Question 1: (157 points)

Templates and Heaps.

Task 01 (117 marks) Create a template class named `heap` with the following specification

1. (2 marks) Private member
 - (a) (1 mark) A one dimensional array of typename `T`.
 - (b) (1 mark) An integer representing size of array.
2. (115) Following public members
 - (a) (25 marks) A member function `insert` with the following specifications
 - i. Input argument 1 of typename `T`
 - ii. Return type `void`
 - iii. Functionality Inserts the given data into the (min) heap
 - iv. (5 marks) Throw exceptions as necessary
 - (b) (20 marks) A member function `MinHeapify` with the following specifications
 - i. Input argument 1 `int` representing array index
 - ii. Return type `void`
 - iii. Functionality Ensures heap properties of subtree rooted at the given input array index.
 - (c) (25 marks) A member function `delete` with the following specifications
 - i. Input argument 1 `void`
 - ii. Return type typename `T`
 - iii. Functionality Deletes the element at the root of the heap and returns its value and preserves min heap properties.
 - iv. (5 marks) Throw exceptions as necessary
 - (d) (10 marks) A member function `search` with the following specifications
 - i. Input argument 1 typename `T`
 - ii. Return type `bool`
 - iii. Functionality Returns true if the element is found in the heap. False otherwise.

- (e) (20 marks) A member function `updateRoot` with the following specifications
 - i. Input argument 1 typename `T`
 - ii. Return type `void`
 - iii. Functionality Updates the root value of the heap with the given input and maintain the heap property.
 - iv. (5 marks) Throw exceptions as necessary
- (f) (10 marks) A member function `printHeap` with the following specifications
 - i. Input argument 1 `void`
 - ii. Return type `void`
 - iii. Functionality Prints the elements of the heap (without changing the heap)

Task 02 (40 marks) Perform the following in the main function. You should ignore all `delete` statement lines in the input file.

1. (10 marks) Read the input file `int-heap.txt`
2. Perform the operations mentioned in the file
3. (10 marks) Read the input file `char-heap.txt`
4. Perform the operations mentioned in the file
5. (10 marks) Read the input file `float-heap.txt`
6. Perform the operations mentioned in the file
7. (10 marks) Read the input file `string-heap.txt`
8. Perform the operations mentioned in the file